ETHICS STATEMENT AND RESEARCH INTEGRITY

Title: A Lightning-Fast Three-Flavor Neutrino Oscillation Calculator in Constant-Density Matter with Built-In Uncertainty Propagation

This document summarizes ethical considerations, compliance statements, and research-integrity practices relevant to the manuscript listed above. It is intended for inclusion as an Ethics / Declarations section in submission packages, institutional repositories, or as supplementary material accompanying the paper.

1. Ethical Approval and Human/Animal Subjects

- This study involves theoretical modeling, algorithm development, and numerical simulations of neutrino oscillation probabilities in constant-density matter.
- No experiments involving human participants or animals were conducted.
- Therefore, no institutional review board (IRB) or animal-care committee approval was required or obtained. This statement is provided in accordance with journal and institutional policies that require explicit declaration when human or animal subjects are not involved.

2. Informed Consent

- The research did not involve human participants, personal data, or identifiable human material. Consequently, no informed-consent procedures were necessary.

3. Data Sources and Use of External Datasets

- Publicly available parameter estimates from the NuFIT global-fitting results were used for benchmark values and uncertainty sampling. NuFIT is cited in the manuscript (I. Esteban et al., JHEP 12, 216 (2024), arXiv:2410.05380). Usage complied with NuFIT's terms of use and citation requirements.
- All other numerical inputs are drawn from published literature or standard physical constants (Particle Data Group). Relevant references are provided in the manuscript.

4. Data Availability and Reproducibility

- All numerical results, scripts, and example input files required to reproduce the principal figures, timing tables, and uncertainty bands will be made available on reasonable request or via a public repository. If a repository URL or DOI is assigned prior to submission (e.g., Zenodo, GitHub with an archived DOI), that link will be included in the final manuscript.
- The repository will include: (a) code implementing exact diagonalization and compact perturbative approximations; (b) the Monte Carlo sampling and Jacobian linearization pipelines; (c) a README with execution instructions and computational environment details (language, major library versions, and required packages); and (d) sample parameter files reproducing the NuFIT-based ensembles used for the uncertainty propagation examples.
- Reproducibility best practices were followed: deterministic random seeds for unit tests, documented numerical tolerances, and unit tests for unitarity and limiting-case checks. Any third-party library dependencies will be listed explicitly in the repository.

5. Code Availability

- The computational code is intended to be released under an open-source license (to be specified by the authors; a permissive license such as MIT or BSD is recommended for maximal reuse). If institutional or collaborator restrictions apply, access terms will be clearly documented and an alternative means of verification (e.g., archived binaries or container images) will be provided.

6. Competing Interests / Conflicts of Interest

- No competing interests or financial conflicts relevant to this manuscript are declared by the authors. Any future or disclosed sources of support that could be perceived as a conflict will be reported transparently in the acknowledgements or a dedicated competing-interests statement.

7. Funding Statement

- No external funding was used to conduct the work reported in this manuscript.
- [If external funding supported any portion of the work, include funder name(s), grant numbers, and a brief description of the funder role here. Replace the previous line with the accurate funding disclosure.]

8. Author Contributions

- Conceptualization and methodological design: A. Chaulagain, A. Dhakal.
- Algorithm design and implementation: A. Chaulagain.
- Uncertainty propagation design and benchmarking: A. Dhakal.
- Validation, figures, and performance testing: A. Chaulagain, A. Dhakal.
- Manuscript drafting and formalization of conclusions: A. Chaulagain, A. Dhakal.
- All listed contributors reviewed the final manuscript and accept responsibility for its content.
- [Adjust author names or roles as required; ensure alignment with journal authorship policies.]

9. Research Integrity and Responsible Conduct

- Computational methods adhere to standard numerical practices (Hermiticity enforcement, symmetrization of Hamiltonians to combat numerical anti-Hermitian artifacts, and unitary-consistency checks).
- All published equations and algorithmic steps reference original sources where applicable (e.g., Minakata & Parke, Akhmedov et al., Particle Data Group) and avoid misattribution.
- Results are reported transparently, including limitations: constant-density approximation, omission of non-standard interactions, and Gaussian approximations for uncertainty propagation in regimes possibly exhibiting non-Gaussian posterior structure.

10. Privacy and Personal Data

- No personal data or identifiable information were used or processed in this research. The manuscript contains no participant-level data.

11. Acknowledgements and Use of Third-Party Tools

- The manuscript acknowledges publicly available resources (NuFIT, PDG) and standard numerical libraries where used. Detailed acknowledgements, including institutional computational resources or informal contributors, will be added to the final manuscript as appropriate.

12. Amendments and Corrections

- Should errors or omissions be discovered post-publication, corrections will be submitted promptly according to the policies of the hosting journal. Code and archived data will be updated with versioning metadata to track any changes.

Contact for Ethics and Data Requests

For questions about research integrity, data access, or reproducibility materials, contact:
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